

Medical Image Processing

Week 11 Exercise - Volume visualization techniques

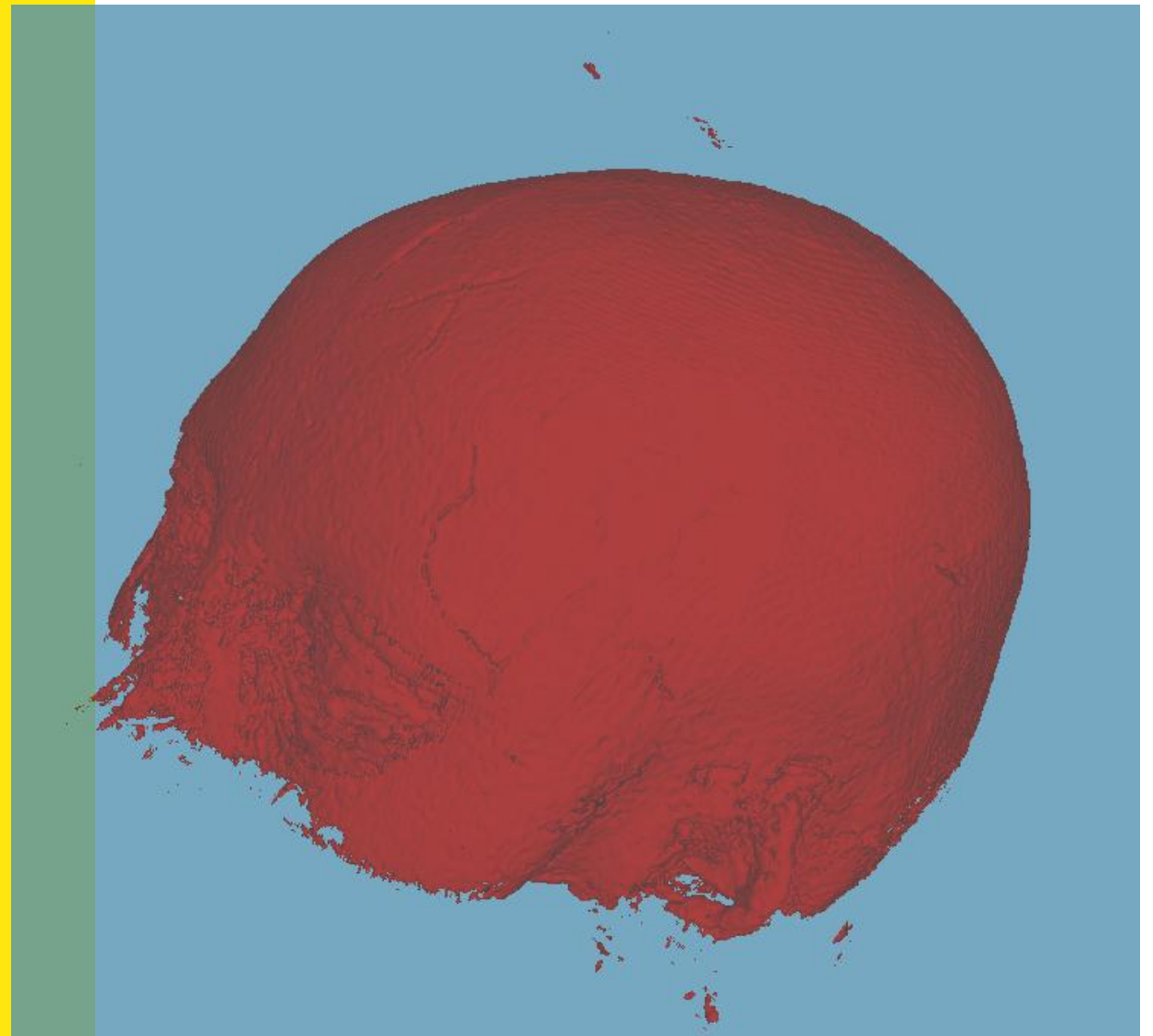
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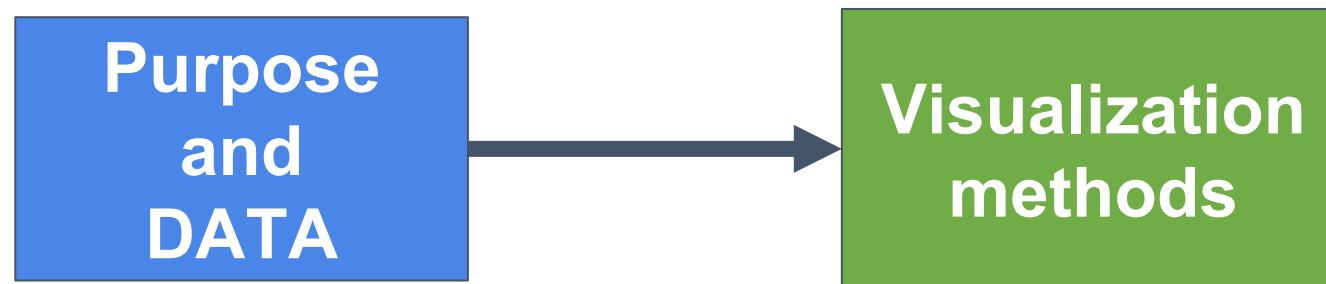
Hatipoglu Osman

Last Updated: December 14th, 2023



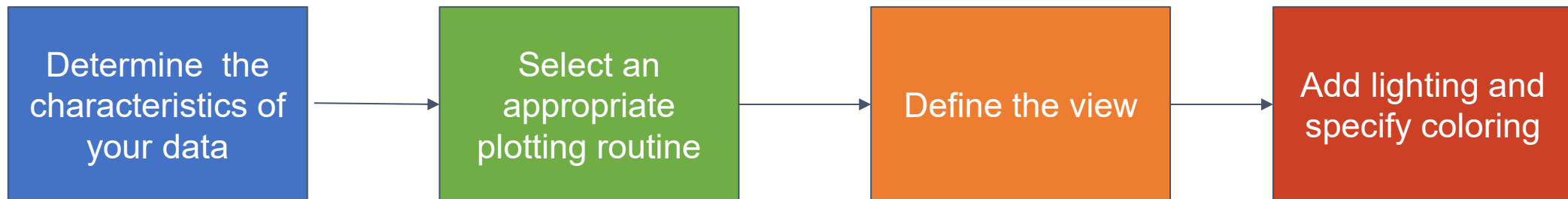
Task description

- **Display** *the surface of the patient in the given **MRI DATA***
- Visualize data using **Isosurface** and **Point Cloud** and **surface rendering**.
- Explain which **pre-processing steps** and parameter choices are required.
- Compare surface rendering visualization with a **volume rendering**.
- **Present the main functions** used in the script.



Volume Visualization

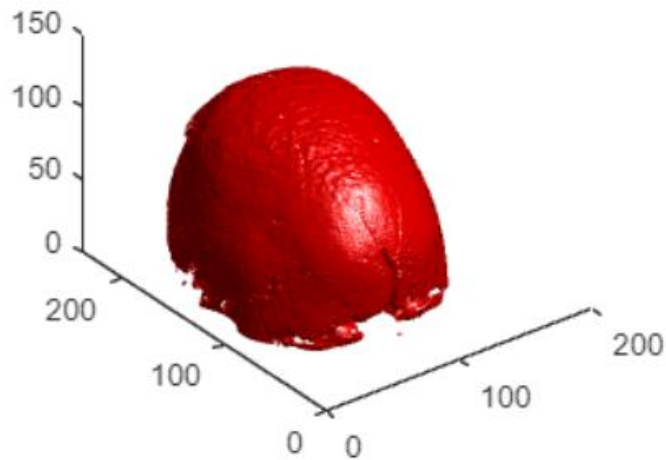
- **Graphical** representations on **3D grids**.
- Two basic type of volume data = **Scalar** or **Vector data**.
- Techniques choice depend on **what you have** and what you **wanna learn**.
- Visualization techniques = **Surface** and **Volume rendering**.



Visualization methods - Display 3-D Medical image Data

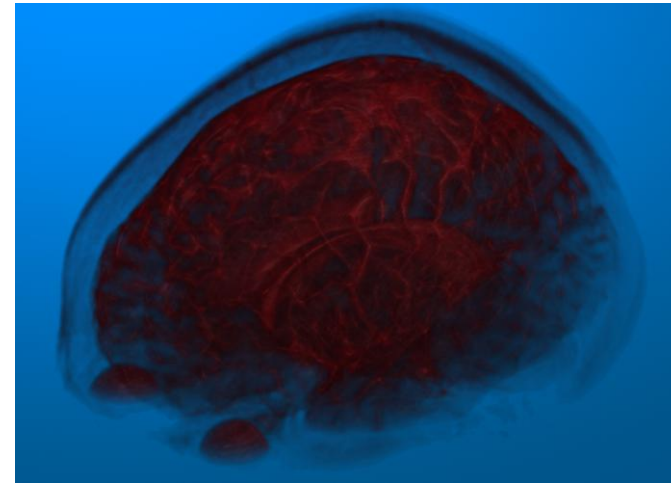
Surface rendering

“... relies on determining the surface in advance. »



Volume rendering

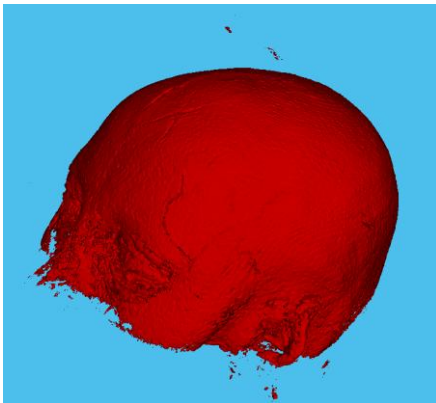
“... classifies relevant info, specifically colors and opacities, and assigns them to voxels based on information about them.”



Computer techniques - functions

Iso-surface

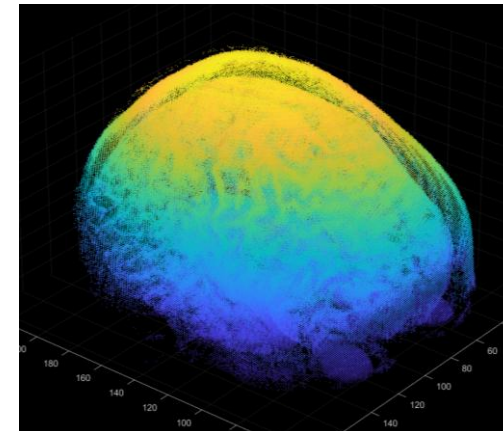
- Meaningful iso-value.
- Polygonal mesh surface.
- Interest in boundary regions.
- Application example: 3D MRI DATA.



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Point cloud

- “Defining a coordinate system (x,y,z).”
- Application example: density 3D CT scan and internal organs and bones.



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Isosurface or Iso-Surface - main functions

– Squeeze the data

```
disp(size(dicomVolume4D));
```

```
% This is a 4D dataset [256 192 1 128], so we have to squeeze it into 3D data
```

```
volume3D = squeeze(dicomVolume4D);
```

– Apply a noise reduction filter

```
% Here we use a median filter which can retain the edge information
```

```
denoiseImage=medfilt3(volume3D,[3,3,3]);
```

– Isosurface

```
[f, v] = isosurface(denoiseImage, isovalue);
```

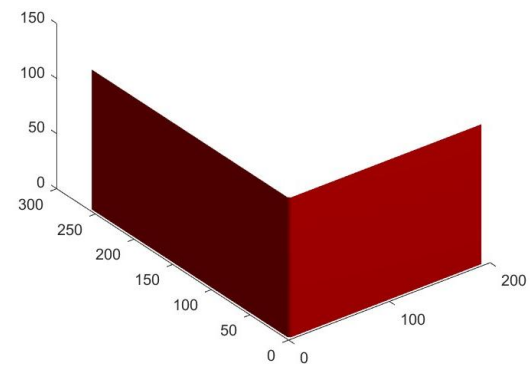
 The isovalue will influence the visualization

```
% "f" refers to face; "v" refers to vertex%
```

```
% use face-vertex meshes to store polygon meshes to achieve surface rendering
```

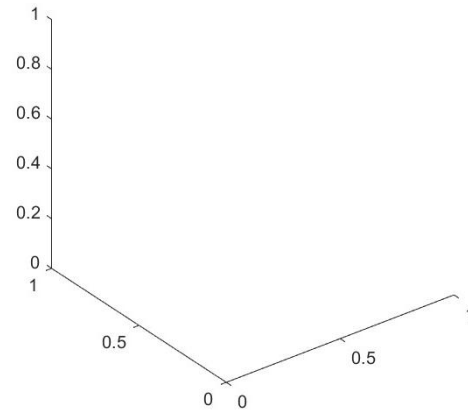
The isovalue will influence the visualization

`isovalue2 = min(denoiseImage(:))`



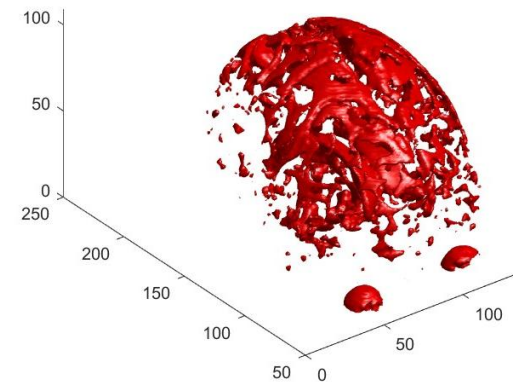
The isovalue is too low

`isovalue3 = max(denoiseImage(:))`



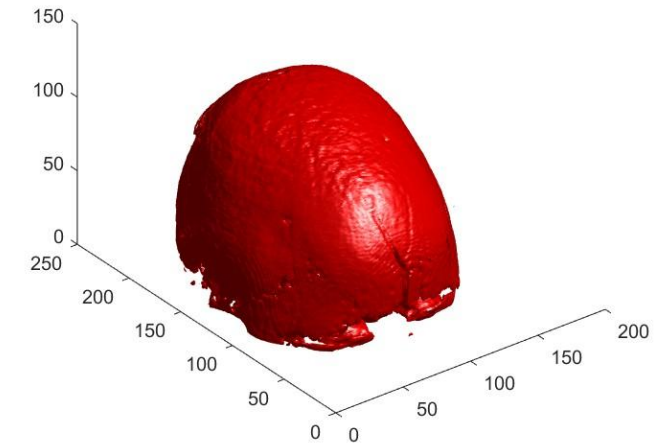
The isovalue is too high

`isovalue4 = 0.5 * max(denoiseImage(:))`



The isovalue is a bit high,
so it can't show the whole structure

`isovalue1 = 0.15 * max(denoiseImage(:))`



The output seems to be better

Point cloud - main functions

- Find points above the threshold to show our target region

```
pointsIdx1 = find(volume3D > threshold1);
```

The region shown varies
with the threshold

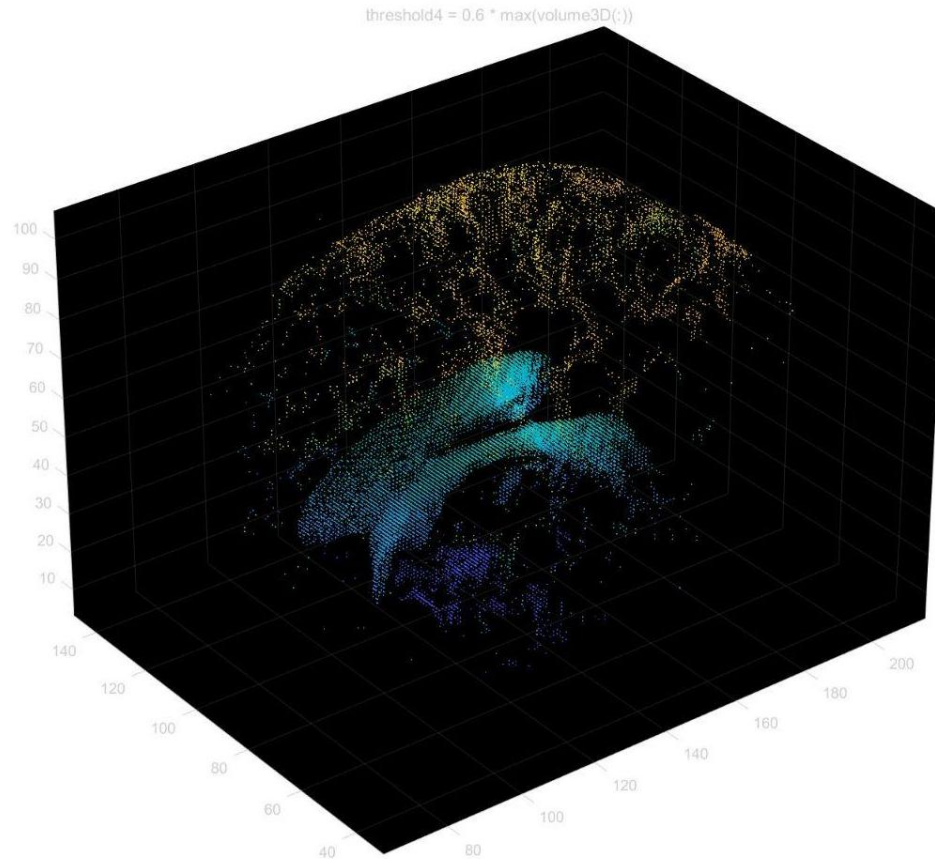
- Convert indices to coordinates of points

```
[x, y, z] = ind2sub(size(volume3D), pointsIdx);
```

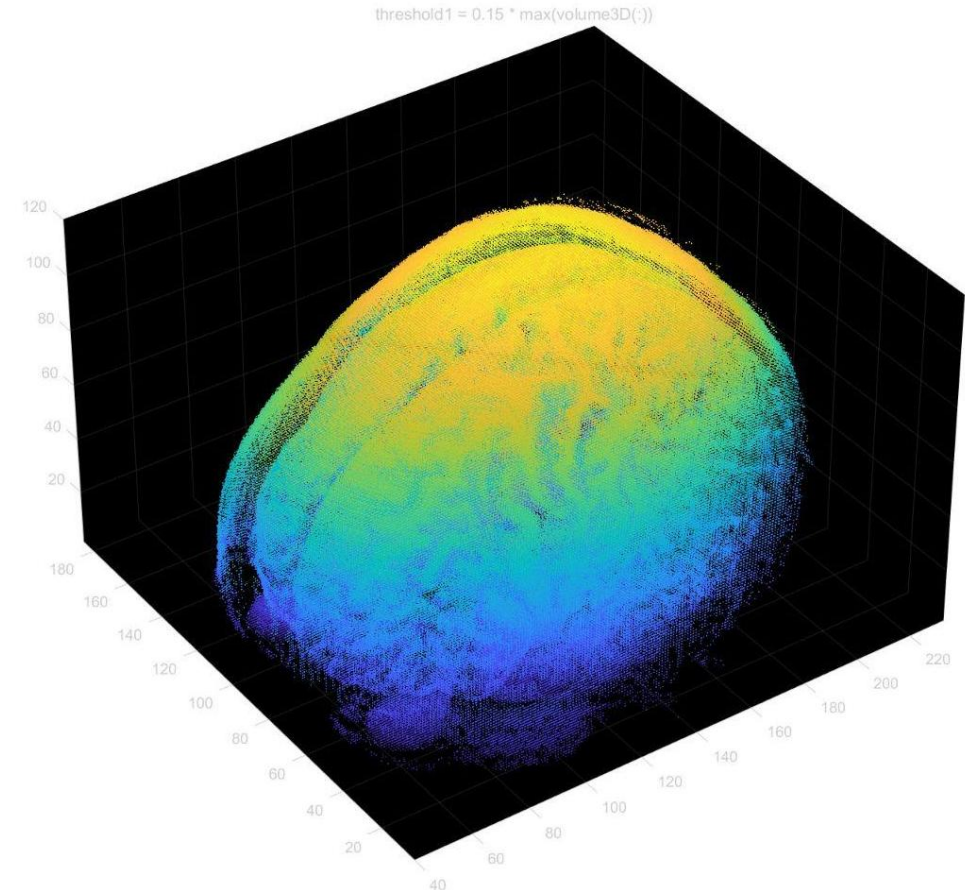
- Create the point cloud using the coordinate dataset

```
ptCloud = pointCloud([x, y, z]);
```


The region shown varies with the threshold



```
"threshold4 = 0.51 * max(volume3D(:))"
```



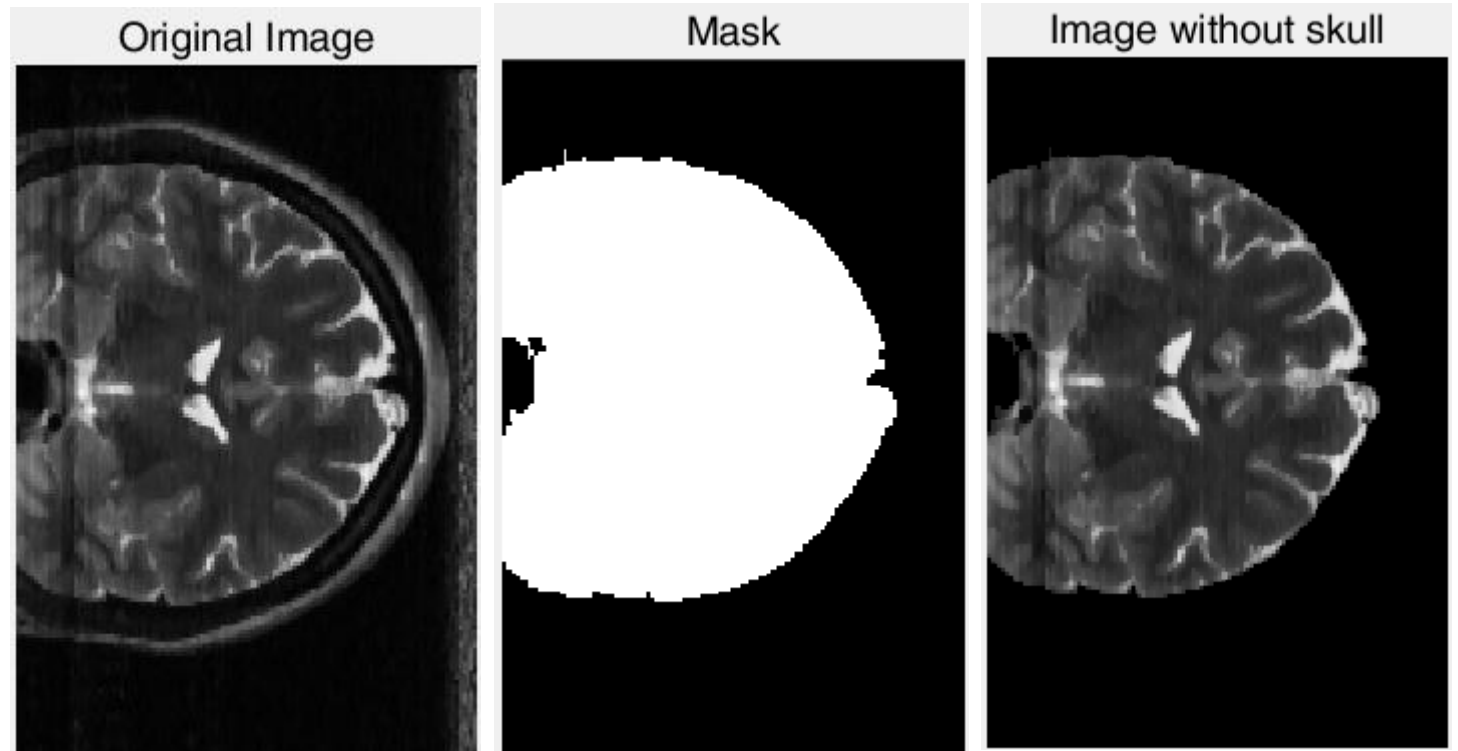
```
"threshold4 = 0.15 * max(volume3D(:))"
```

Then we use some additional preprocessing steps to
remove the skull to directly view the cortex.

Main principle of removing skull

- get **mask** of interested area
- multiple mask and original image

```
% use circulation to process each slide, N is the number of slides
for n = 1:N
    switch position
        case 1
            Original_data = DCM(:,n,:);
        case 2
            Original_data = DCM(n,:,:);
    end
    Original_image = squeeze(Original_data); % convert 3D array to 2D
    Binary_image = Original_image > threshold; % threshold segmentation
    % get maximum connected domain
    imLabel = bwlabel(Binary_image);
    stats = regionprops(imLabel,'Area');
    [b,index]=sort([stats.Area],'descend');
    % 1 means only get the first maximum
    if length(stats)<1
        Mask = imLabel;
    else
        Mask = ismember(imLabel,index(1:1)); % get original mask
    end
    % remove the details of the mask to get the complete image
    Mask = imcomplement(Mask); % get reverse image
    Mask = bwareadopen(Mask, 100); % remove patterns smaller than 100 pixels
    Mask = imcomplement(Mask); % get reverse image
    % remove obvious useless information on the top of the image
    if Mask((x+y)/4,z-1)==1
        Mask = zeros;
    end
    % multiple the original image and mask to get interested area
    New_image= bsxfun(@times, Original_image, cast(Mask,class(Original_image)));
    New_surface = reshape(New_image, array);
    % assign value to New_dicom
    switch position
        case 1
            New_dicom(:,n,:) = New_surface;
        case 2
            New_dicom(n,:,:)= New_surface;
    end
end
```



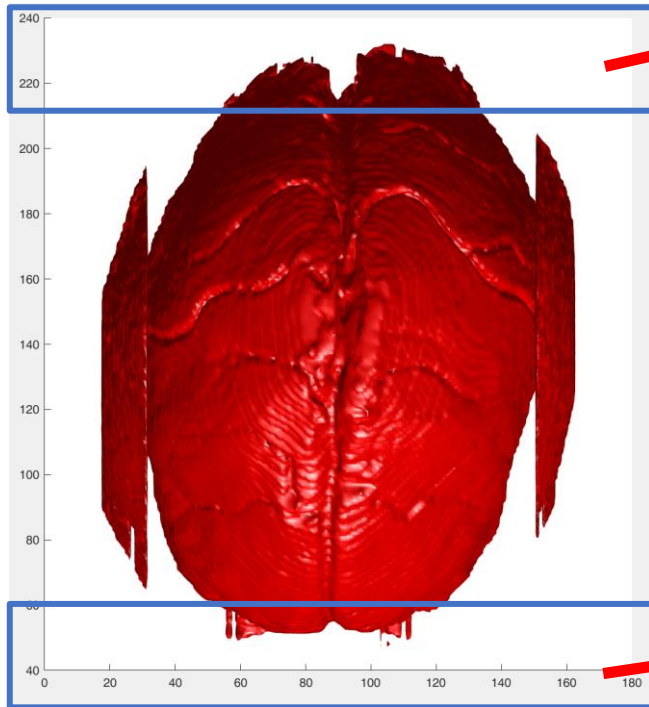
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```
% use function to remove skull in xz and yz surface
DCM_xz = My_removeskull(DCM, 'xz', 0.13);
DCM_yz = My_removeskull(DCM, 'yz', 0.13);

% replace some wrong data of DCM_yz with DCM_xz
for m = 1:60
    DCM_yz(m, :, :) = DCM_xz(m, :, :);
end
for m = 210:256
    DCM_yz(m, :, :) = DCM_xz(m, :, :);
end
```

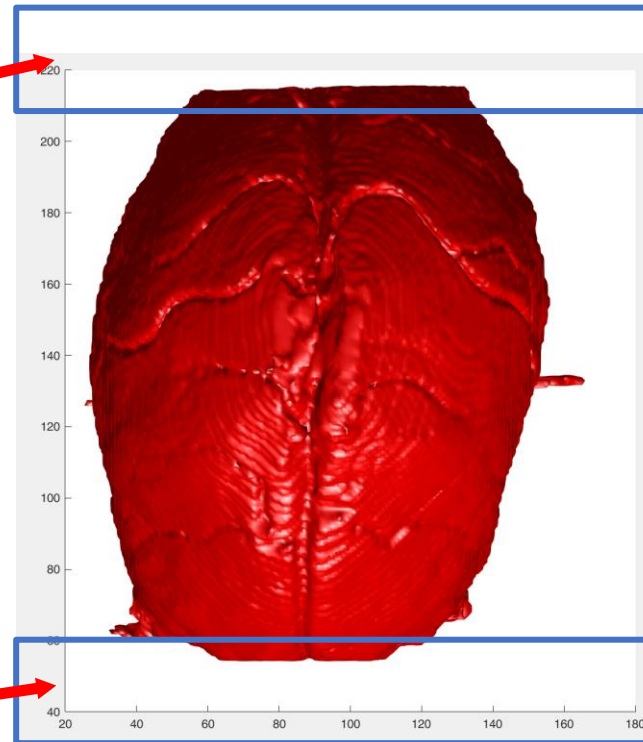
Main principle of removing skull

process by **xz** surface



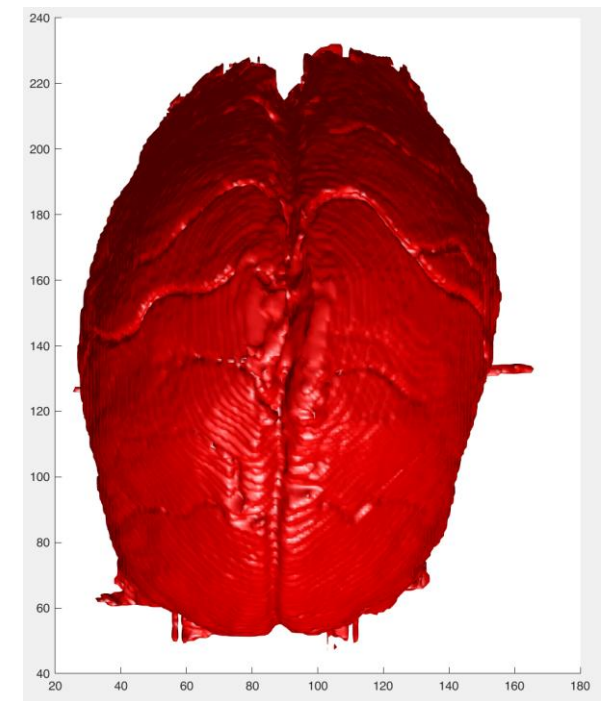
Matlab, 2023

process by **yz** surface



Matlab, 2023

combine them

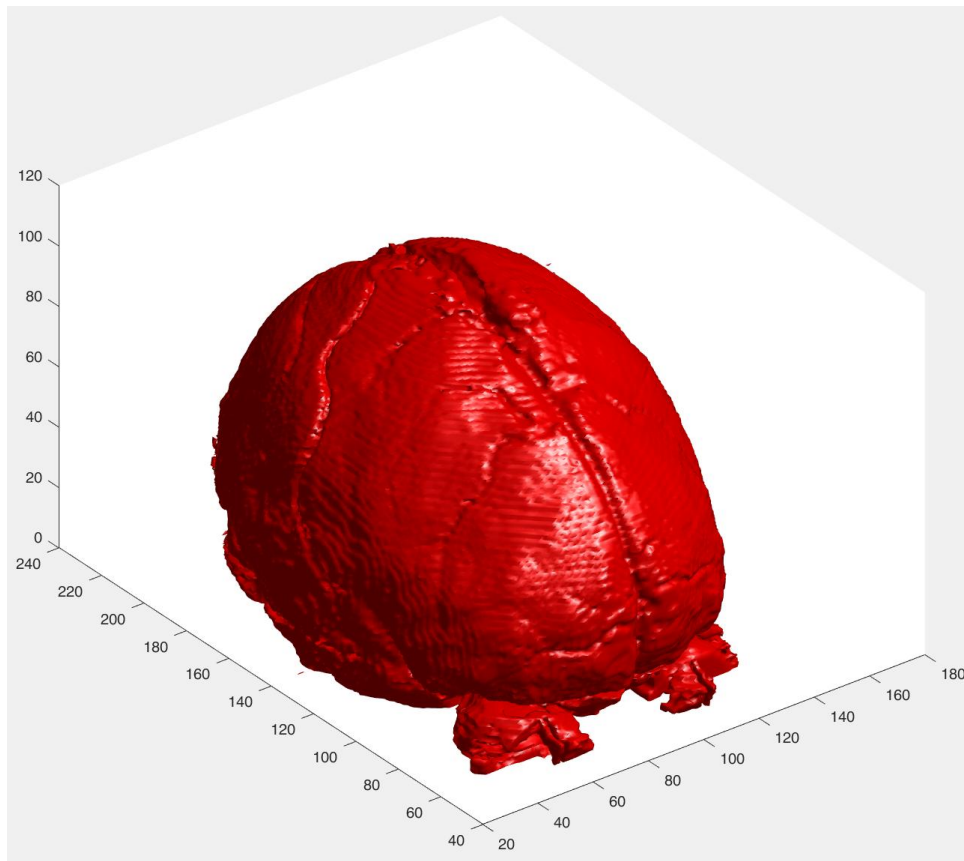


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replace data of yz result with xz result in two areas

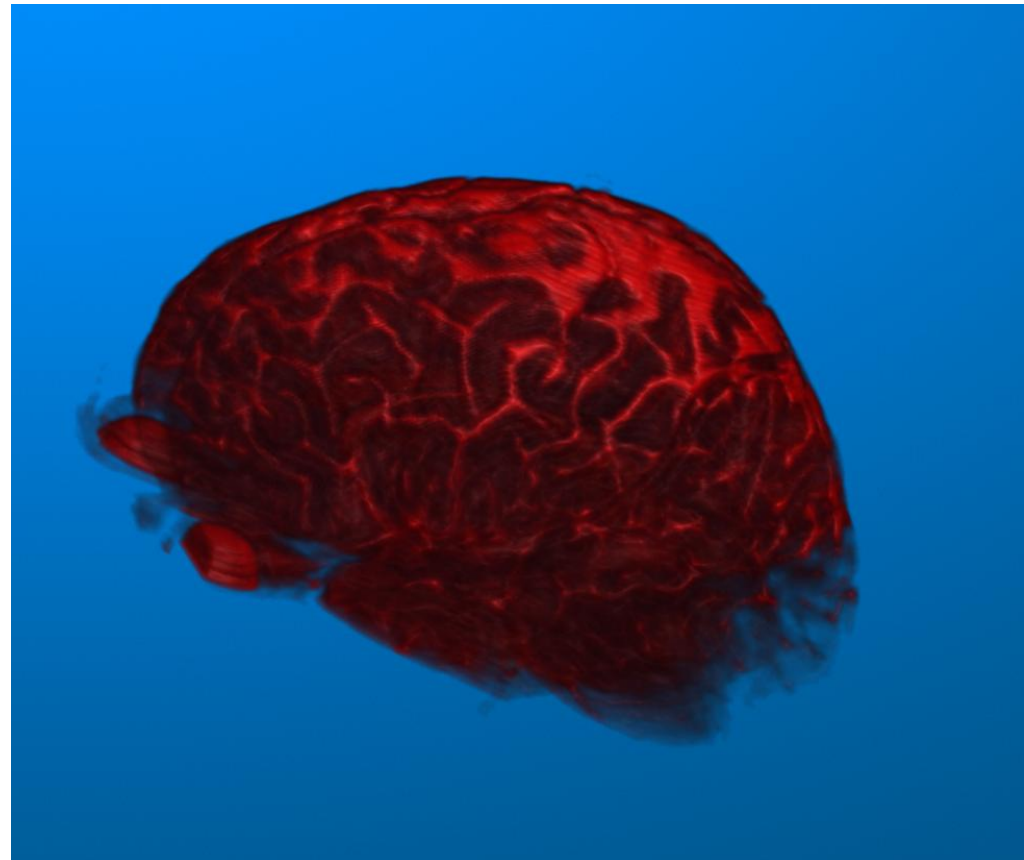
Results without skull

Surface Rendering



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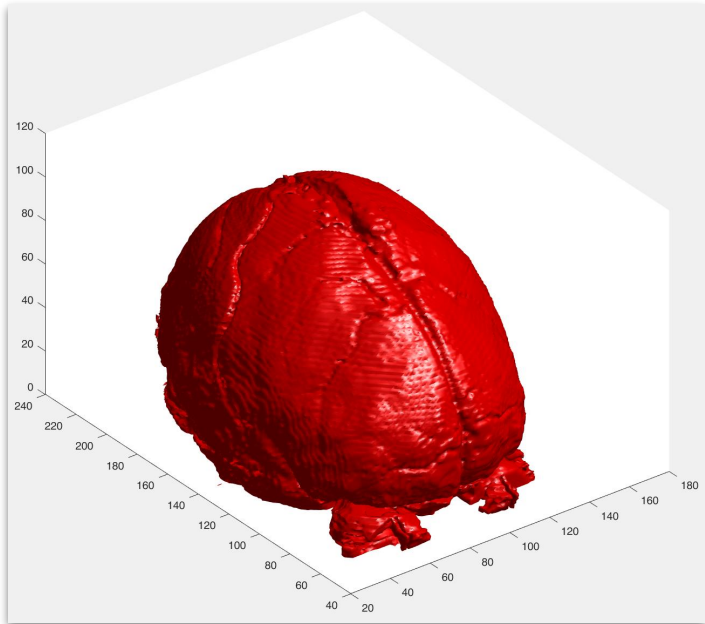
Volume rendering



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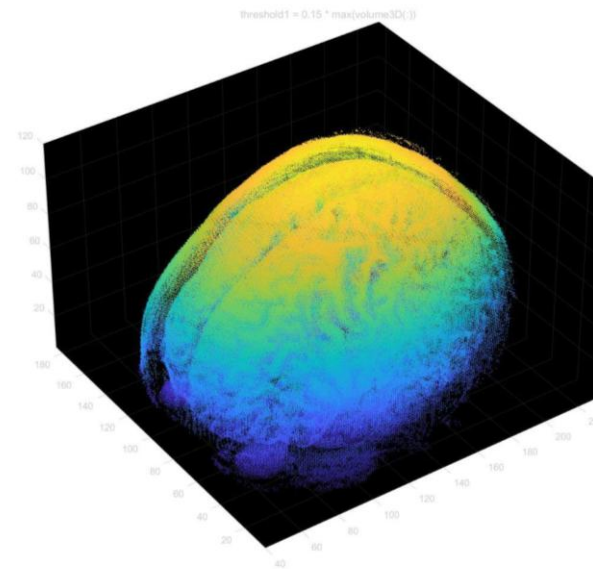
Comparison between views

Isosurface - Surface rendering

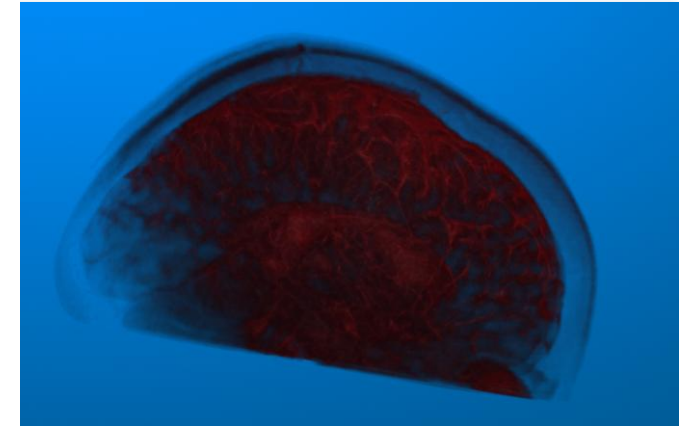


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Point Cloud - Volume rendering



Matlab, 2023



Suggestion exercises for modifications

Isosurface:

- Try to use Gaussian filter to better visualization instead Median filter.
- Try to use Dual contouring instead of Marching Cube algorithm (sharp edges and corners).
- Think in a way to remove the skull completely.



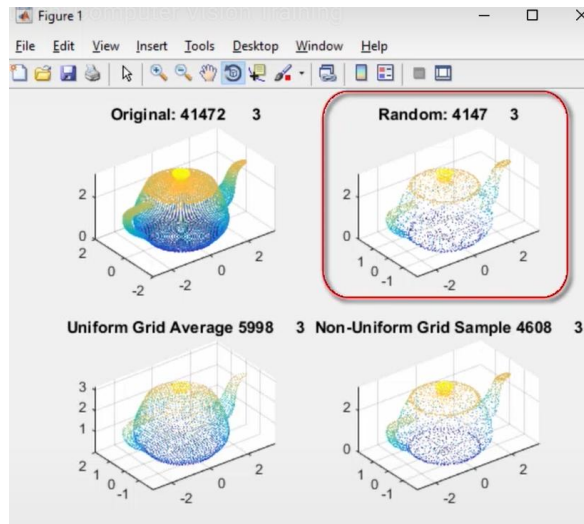
Plus:

- Active Contour Models(snakes)
- Topological methods
- Machine Learning approaches
 - Deep Learning
 - 3D and Volumetric CNN

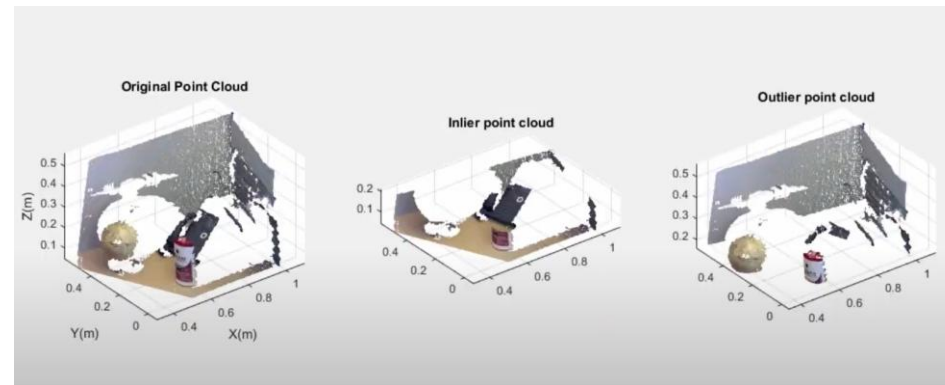
Suggestion exercises for modifications

Point Cloud:

- Try to use Downsampling (Random, ... Non-Uniform Grid Sample - the best)
- Try to use Extraction ROI (Outlier this time).



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References

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- Matlab. Point Cloud. <https://ch.mathworks.com/help/vision/ref/pointcloud.html>. Accessed on 12 December 2023.
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